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Antiferromagnetic Spin Fluctuations and Superconductivity in Two-Dimensional Metals —A Possible Model for High T_c Oxides

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Spin fluctuations in antiferromagnetic and nearly antiferromagnetic two dimensional (square lattice) itinerant electron systems, as a possible model for high T_c superconductors, are investigated by using the self-consistent renormalization theory. The electrical resistivity and the nuclear spin relaxation rate due to the spin fluctuations in the normal state are calculated. The results in the nearly antiferromagnetic regime as applied to high T_c oxides seem to explain the experimental results both in their temperature dependence and in their orders of magnitudes. By using the same spin fluctuations we discuss superconductivity due to the spin fluctuation mechanism within a weak coupling theory. The order parameter is shown to have B_{1g} or A_{2g} symmetry and the critical temperature is evaluated to be of the right order of magnitude.

[spin fluctuation, high T_c superconductivity, electrical resistivity, nuclear spin relaxation, specific heat, theory]

§1. Introduction

The mechanism of superconductivity in high T_c oxides has been a subject of current interest.[1] Although a number of proposals have been presented so far, the most striking features arising from experimental studies seem to be the two-dimensional (2D) character of electrons and strong antiferromagnetic spin fluctuations in them.

Theoretical approaches so far presented are mostly based on the strong correlation picture. This picture is favored firstly because the mother substances of high T_c oxides are antiferromagnetic insulators where the approach from the strong correlation limit is established.[2] Secondly, several experimental studies suggest that the antiferromagnetic spin fluctuations in these insulator oxides persist when they are doped with holes or electrons so that they become superconductive. Thirdly, the temperature dependence of the electrical resistivity above T_c in these high T_c oxides is linearly proportional to temperature in a wide range of temperature, quite unusual behavior for normal metals or for Fermi liquids. Convincing explanation for this linear temperature dependence of the electrical resistivity, however, does not seem to have been presented so far.

On the other hand, the approach from the side of weaker correlation should also be useful firstly because the high T_c oxides when doped further seem to become normal metals as many experimental investigations indicate. Secondly it is conceivable that as one dopes a Mott insulator with increasing number of carriers it very rapidly becomes metallic due to rapid increase of the mutual screening effects of carriers and the screening of impurity potentials. Thirdly, the spin fluctuations are expected to change rather gradually as one goes from an insulator phase to a metallic phase by reducing effective interactions,[3] thus the observed persistence of large antiferromagnetic fluctuations with doping may not necessarily mean the persistence of local moments or the strong correlation limit.

The purpose of the present paper is first to discuss various physical properties of

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two-dimensional itinerant electron antiferromagnets, as a possible model of high T_c oxides, on the basis of the self-consistent renormalization (SCR) theory of spin fluctuations which has been extremely successful for 3-dimensional weak itinerant ferro- and antiferromagnetism,[4] and then to apply the results to discuss superconductivity. We first discuss in §2 spin fluctuations and the staggered susceptibility in 2D-itinerant antiferromagnets and then calculate electrical resistivity in §3, nuclear spin relaxation rate in §4, specific heat in §5 and finally discuss in §6 superconductivity due to the spin fluctuation mechanism. An interesting feature of this approach is that one can make quantitative comparisons between theory and experiment through relatively small number of parameters which can in principle be evaluated from experimental results.

As results, we will see that the linear temperature dependence of the electrical resistivity and the nuclear spin relaxation rate are rather well explained including their orders of magnitude within the SCR theory for nearly antiferromagnetic regime. The spin fluctuation mechanism for the superconductivity gives rise to the order parameter of B_{1g} or A_{2g} symmetry and the estimated critical temperature is of the right order of magnitude. Finally in §7 we conclude with summary and discussions.

§2. Spin Fluctuations and Staggered Susceptibility

According to the SCR theory for itinerant antiferromagnetism,[5,6] we have the following magnetic equation of state:

$$\frac{1-\alpha_s}{\chi_0(\boldsymbol{Q})}+\frac{5}{3}F_s S_L^2+F_s M_s^2=\frac{h_s}{M_s}, \tag{2.1}$$

where M_s is the staggered magnetization (per magnetic atom), $h_s=2\mu_B H_s$ the staggered magnetic field in energy units, $\chi_0(\boldsymbol{Q})$ the staggered susceptibility without the exchange enhancement, $\alpha_s=2I\chi_0(\boldsymbol{Q})$, I being the intra-atomic exchange interaction (we take a model of local exchange interaction), F_s the mode-mode coupling constant for the Fourier components of the spin density around the antiferromagnetic wave vector $\boldsymbol{Q}$, and S_L^2 is the mean square local amplitude of the spin fluctuation given by

$$\frac{1}{3}S_L^2=\frac{1}{3}S_{L0}^2+\frac{2}{\pi}\int_0^\infty d\omega\frac{1}{e^{\omega/T}-1}\sum_q \times \mathrm{Im}\,\chi(\boldsymbol{Q}+\boldsymbol{q},\omega), \tag{2.2}$$

S_{L0}^2 being the zero point contribution and $\chi(\boldsymbol{Q}+\boldsymbol{q},\omega)$ the dynamical susceptibility. We take energy units for ω and T, so that $\hbar=1$, $k_B=1$. The second term on the left hand side of eq. (2.1) represents the effect of spin fluctuations which is not contained in the Hartree-Fock-RPA theory. The dynamical susceptibility around the staggered component is given by

$$\chi(\boldsymbol{Q}+\boldsymbol{q},\omega)=\chi(\boldsymbol{Q}+\boldsymbol{q})/(1-i\omega/\Gamma_{Q+q}),$$

with

$$\Gamma_{Q+q}=\Gamma_0(\kappa_s^2+q^2)=\Gamma_0/2\alpha_s\bar{A}\chi(\boldsymbol{Q}+\boldsymbol{q}),$$
$$\Gamma_0=A/C,\quad \bar{A}=A/2\chi_0(\boldsymbol{Q}), \tag{2.3}$$
$$\kappa_s^2=\chi_0(\boldsymbol{Q})/\alpha_s A\chi(\boldsymbol{Q}),$$

where κ_s is the inverse antiferromagnetic correlation length and the parameters A and C are defined by the following expansion form for small $\boldsymbol{q}$ and small ω of the dynamical susceptibility of non-interacting electrons:

$$\chi_0(\boldsymbol{Q}+\boldsymbol{q},\omega)=\chi_0(\boldsymbol{Q})\{1-Aq^2+iC\omega+\cdots\}.$$

For a 2D-system we have a magnetic order at $T=0$ when

$$M_s^2(0)=\frac{\alpha_s-1}{F_s\chi_0(\boldsymbol{Q})}-\frac{5}{3}S_{L0}^2(0) \tag{2.4}$$

has a positive value, as is seen from eq. (2.1).

At finite temperatures we calculate eq. (2.2) explicitly by using eq. (2.3) and get the following result

$$\frac{1}{3}S_{LT}^2=\frac{1}{3}(S_L^2-S_{L0}^2)$$
$$=\frac{(1+\delta_s)}{\alpha_s\bar{A}q_B^2}\frac{1}{\pi}\int_0^\infty d\omega\frac{1}{e^{\omega/T}-1}\left\{\tan^{-1}\frac{\omega}{\Gamma_0\kappa_s^2}-\tan^{-1}\frac{\omega}{\Gamma_0(\kappa_s^2+q_B^2)}\right\}, \tag{2.5}$$

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with $\delta_s=A\kappa_s^2$, where q_B is the cut off wave vector and here we take

$$s_0 q_B^2/4\pi=1, \tag{2.6}$$

s_0 being the area per magnetic atom in the 2D plane.

We first note that there is no magnetic order at finite temperatures. Since we have $\kappa_s^2=0$ in the ordered state, eq. (2.5) reduces to

$$\frac{1}{3}S_{LT}^2=\frac{1}{\alpha_s\bar{A}q_B^2}\frac{1}{\pi}\int_0^\infty d\omega\,\frac{1}{e^{\omega/T}-1}\tan^{-1}\frac{\Gamma_0 q_B^2}{\omega}\to\infty \quad \text{for } T>0. \tag{2.7}$$

Thus at finite temperatures we need to calculate the staggered susceptibility $\chi(\boldsymbol{Q})=M_s/h_s$ from eqs. (2.1) and (2.5). As a matter of fact the integrals in the expression for S_{LT}^2 can be performed and we get the following equation for reduced inverse susceptibility y or the squared correlation length in units of q_B:

$$y=y_0+\frac{1}{2}y_1 t(1+\delta_s)\{\phi(y/t)-\phi(y/t+1/t)\},$$

with

$$y=\kappa_s^2/q_B^2,\ \delta_s=A\kappa_s^2,$$
$$y_0=(1-\alpha_s)/2\alpha_s\chi_0(\boldsymbol{Q})T_A+(5F_s/6\alpha_s T_A)S_{L0}^2,$$
$$y_1=5F_s T_0/T_A^2,$$
$$t=T/T_0,\ T_0=\Gamma_0 q_B^2/2\pi,\ T_A=\bar{A}q_B^2,$$

and

$$\phi(x)=-\left(x-\frac{1}{2}\right)\log x+x+\log\Gamma(x)-\log\sqrt{2\pi}. \tag{2.8}$$

We first solve this equation numerically and show the result in Fig. 1 where we plot y against reduced temperature t for $y_1=3$ and for various values of y_0.

For $y_0<0$ we have magnetic order at $T=0$, and eq. (2.8) for small t reduces to

$$y\sim t\exp(-4|y_0|/y_1 t). \tag{2.9}$$

On the other hand for $y_0>0$ we have $y-y_0\propto t^2$

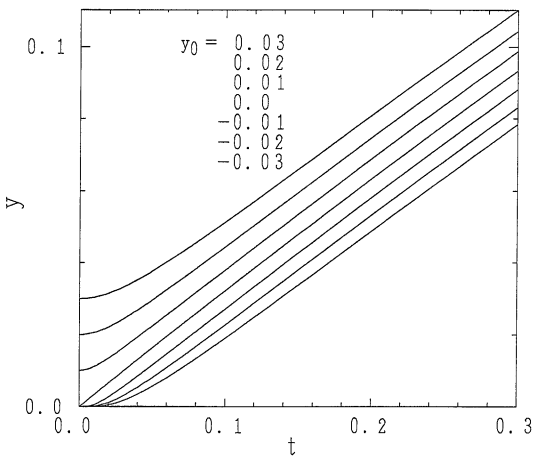


Fig. 1. Inversed staggered susceptibility (reduced) or inversed squared correlation length in reciprocal lattice units against reduced temperature for $y_1=3$.

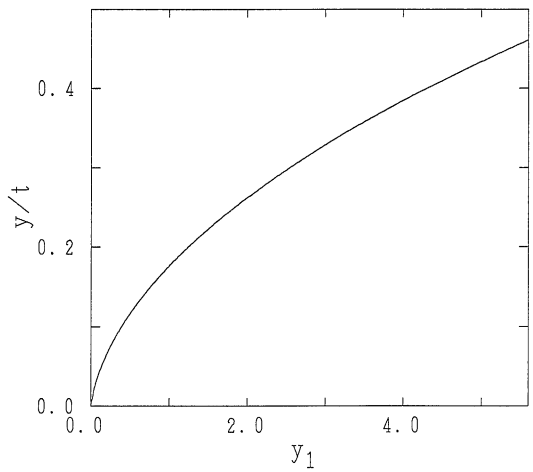


Fig. 2. Inversed Curie constant in units of $T_0/2\alpha_s T_A$: y/t vs. y_1 for $y_0=0$.

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for small t.

At elevated temperatures y is linear in t, i.e., the Curie-Weiss law holds for both $y_0<0$ and $y_0>0$. The Curie constant or the slope of the $y-t$ plot is approximately constant for varying values of y_0. Thus its dependence on y_1 may be seen from Fig. 2 where we plot y/t against y_1 for $y_0=0$.

In the following sections we discuss various physical properties related with the spin fluctuations. Here we rewrite the dynamical susceptibility in terms of the parameters defined in eq. (2.8).

$$\mathrm{Im}\,\chi(\boldsymbol{Q}+\boldsymbol{q},\omega)=\frac{\pi T_0}{\alpha_s T_A}\frac{\omega}{\omega^2+\{2\pi T_0[y+(q/q_B)^2]\}^2}. \tag{2.10}$$

The spin fluctuations can be observed directly by neutron scattering experiments. In particular one can measure the staggered susceptibility or the inverse antiferromagnetic correlation length $\sqrt{y}$ (in units of q_B) and determine the values for the parameters T_0 and T_A, which characterize the spin fluctuations. Neutron scattering studies of high T_c and related oxides have been performed intensively in recent years[7] and two dimensional character of spin fluctuations was confirmed. These materials consist of layers of CuO_2 and in each layer Cu-atoms form a square lattice. The neutron scattering intensity peaks around $(\pi/a, \pi/a, 0)$.

For $La_{2-x}Sr_xCuO_4$, reported correlation lengths in the high T_c range $(0.1<x<0.18)$ are about 10 Å at low temperatures and its temperature dependence is rather weak. Thus we may estimate $y_0\sim 0.01$. On the other hand for the mother antiferromagnet La_2CuO_4 and doped semiconductors (small x) the correlation lengths are quite large and decreases exponentially with increasing temperature as predicted for 2D Heisenberg antiferromagnet[8] and here also by eq. (2.9). This suggests that y_0 is negative for small x and increases with increasing x, changing sign before superconductivity sets in.

As for the temperature dependence of y in high T_c materials, the reported measurements do not seem to have enough accuracy to make comparison with Fig. 1. Estimation of the parameters T_0 and T_A from reported measurements seems to be hard except for very crude order estimations as will be discussed below. More detailed measurements are desired in future.

§3. Electrical Resistivity Due to Spin Fluctuations

The electrical resistivity due to spin fluctuations is given by the following expression.[9,10]

$$R=\frac{6\hbar J^2}{N_A e^2 n^2 \pi T}\frac{1}{N_A}\sum_q\int_{-\infty}^{\infty} d\omega q_x^2 n(\omega)[n(\omega)+1]\chi_c''(q,\omega)\,\mathrm{Im}\,\chi(q,\omega), \tag{3.1}$$

with

$$\chi_c''(q,\omega)=\pi\sum_k \delta(\varepsilon_k-\varepsilon_{k-q}-\omega)[f(\varepsilon_{k-q})-f(\varepsilon_k)],$$

where J is the exchange coupling constant between the conduction electron and the spin fluctuation, N_A the number of magnetic atoms per unit volume, n the number of electrons per atom, ε_k the energy dispersion of conduction electrons and $n(\omega)$ and $f(\varepsilon)$ are the Bose and Fermi distribution functions, respectively. For the 2D system, using the results of the preceding section, we calculate eq. (3.1) as follows:

$$\begin{aligned} R(T)&=r\bar{R}(T),\\ \bar{R}(T)&=\frac{1}{T}\frac{1}{N_A}\sum_q\int_{-\infty}^{\infty} d\omega \frac{e^{\omega/T}}{(e^{\omega/T}-1)^2}\frac{\omega^2}{\omega^2+\{2\pi T_0[y+(q/q_B)^2]\}^2},\\ r&=\frac{6\hbar Q^2 J^2\chi_c(\boldsymbol{Q})\chi_0(\boldsymbol{Q})(C_0/C)}{N_A e^2\pi\alpha_s n^2}, \end{aligned} \tag{3.2}$$

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where we used

$$\chi_c''(\boldsymbol{Q}+\boldsymbol{q}, \omega)\simeq\chi_c(\boldsymbol{Q})C_0\omega, \tag{3.3}$$

for small $\boldsymbol{q}$ and small ω. Now the calculation of $\bar{R}(T)$ is straightforward and we get

$$\begin{aligned}\bar{R}(T)&=\frac{t}{\pi}\int_0^\infty dx\frac{x\,e^x}{(e^x-1)^2}\left\{\tan^{-1}\frac{tx}{2\pi y}-\tan^{-1}\frac{tx}{2\pi(y+1)}\right\}\\&=y\left\{\log\frac{y}{t}-\psi\left(\frac{y}{t}\right)\right\}-(y+1)\left\{\log\frac{y+1}{t}-\psi\left(\frac{y+1}{t}\right)\right\}+t\left\{\phi\left(\frac{y}{t}\right)-\phi\left(\frac{y+1}{t}\right)\right\},\end{aligned} \tag{3.4}$$

where $\psi(z)$ is the digamma function and $\phi(z)$ is given by eq. (2.8).

From eq. (3.4) one can immediately see that $\bar{R}(T)$ is linearly proportional to T when y/t is constant. Thus from the result of the preceding section we expect good linearity in T of the electrical resistivity in a wide range of temperature. This is clearly seen in Fig. 3 where we plot $\bar{R}(T)$ against t as calculated from eq. (3.4) for various values of y_0 for $y_1=3$. One of the unique features of high T_c superconductor oxides is the linear temperature dependence of their electrical resistivities above T_c in a wide range of temperature. We now wish to compare the present result with experiment on high T_c oxides.

We now assume that in these oxides the spin fluctuations are produced by the electrons as charge carriers. In other words, we take a one fluid picture for electrons. Then we have $C_0=C$, $2J\chi_0(\boldsymbol{Q})=2J\chi_c(\boldsymbol{Q})=\alpha_s$, and furthermore the factor of 4 arises in the expression for $R(T)$ or r since the exchange field is now twice as large as the case where the carriers and scatterers are considered to be different. Our model for these oxides consists of layers of square lattices with a lattice constant a, separated by a lattice constant c. Then we have

$$r=\frac{6\pi\hbar\alpha_s}{e^2n^2}\left(\frac{Qa}{\pi}\right)^2 c. \tag{3.5}$$

The antiferromagnetic wave vector is $\boldsymbol{Q}\simeq(\pi/a, \pi/a, 0)$ and we have $c=6.6$ Å (LaSrCuO), 5.8 Å (YBaCuO), 6.1 Å (NdCeCuO), etc. Thus assuming $\alpha_s\simeq 1$, $n\simeq 1$ we get

$$R(T)\simeq 1\times 10^{-2}\bar{R}(T)\ \Omega\mathrm{cm}. \tag{3.6}$$

The observed values for the slope of the resistivity are[11,12] $\sim 2\times 10^{-6}\ \Omega\mathrm{cm}\,K^{-1}$ (LaSrCuO, $T_c=40$), $\sim 1\times 10^{-6}\ \Omega\mathrm{cm\ K^{-1}}$ (YBaCuO, $T_c=90$), $\sim 2.5\times 10^{-6}\ \Omega\mathrm{cm\ K^{-1}}$ (NdCeCuO, $T_c=22.5$), etc. From Fig. 3 we find $d\bar{R}(T)/dT\sim 0.4/T_0$ for $y_1=3$, $y_0=0.01$. This value is rather insensitive to the value of y_0 or y_1 as will be seen from Fig. 2 and Fig. 4 where we show $\bar{R}(T)/t$ vs y/t for $y_0=0$, $t\to 0$. Thus we estimate roughly

$$T_0\sim 1600-4000\ \mathrm{K},$$

for these materials. These values are not only in a reasonable range of the parameters but also consistently explain the observed results for the nuclear spin relaxation rate as will be discussed in the following section. A very crude estimate of T_0 for $YBa_2Cu_3O_{6.5}$($T_c=50$ K) is possible from recent neutron scattering experiment.[13] According to the report we have $\Gamma_Q=2\pi T_0 y\sim 30$ meV at low temperatures. Thus assuming $y_0\sim 0.01$ we estimate $T_0\sim 5000$ K. For $YBa_2Cu_3O_{6.45}$($T_c=45$ K), however, the corresponding value, as estimated from the

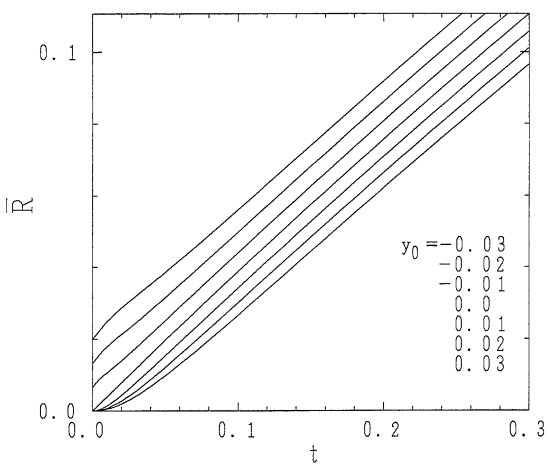


Fig. 3. Reduced electrical resistivity against reduced temperature for $y_1=3$.

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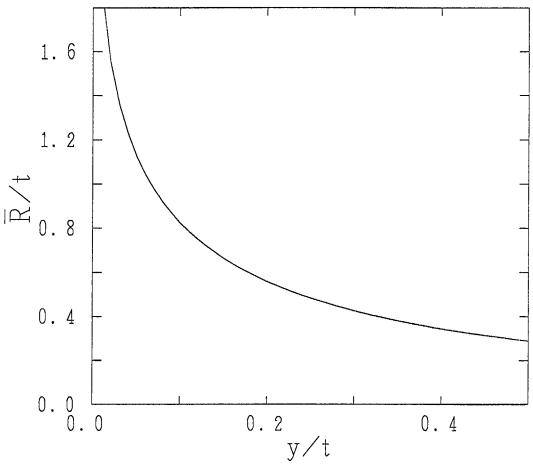


Fig. 4. Linear coefficient of reduced temperature in the reduced electrical resistivity: $\bar{R}(t)/t$ against y/t for $y_0=0$ and $t\to 0$.

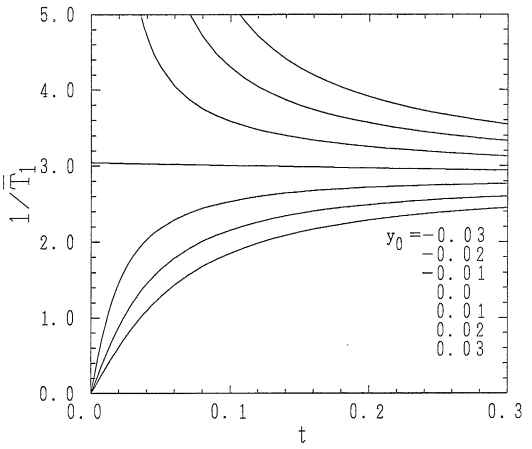


Fig. 5. Reduced nuclear spin-lattice relaxation rate against reduced temperature for $y_1=3$.

reported value $\Gamma_Q\sim 2.5$ meV, becomes $T_0=460$ K. Although more convincing measurements for various substances are desirable, our value for T_0 seems to have a fair order of magnitude.

In some high T_c oxides with relatively low T_c the resistivity obeys the T^2 law at low temperatures and gradually becomes linear in T with increasing temperature.[12] This behavior is also consistent with the present results in the nearly antiferromagnetic regime ($y_0>0$) as is visualized in Fig. 3 for relatively large values of $y_0>0$.

§4. Nuclear Spin Relaxation Rate

Nuclear spin-lattice relaxation rate $1/T_1$ is generally given by[14]

$$\frac{1}{T_1}=\gamma_N^2 T N_A^{-1}\sum_q \frac{|A_q|^2\,\mathrm{Im}\,\chi^{-+}(q,\omega_0)}{\omega_0}, \qquad (4.1)$$

where A_q is the coupling constant for the hyperfine interaction between the nuclear spin and the q-component of the spin density and ω_0 is the nuclear resonance frequency.

We consider here only the nuclei of magnetic atoms where the q-dependence of A_q may approximately be neglected. By using the result of §2 in eq. (4.1) and taking the limit of $\omega_0\to 0$, we get

$$(1/T_1)_s=2\gamma_N^2 A_{hf}^2 T N_A^{-1}\sum_q \chi(Q+q)/\Gamma_{Q+q}$$
$$=\frac{\hbar\gamma_N^2 A_{hf}^2}{2\pi\alpha_s T_A}\frac{1}{\bar{T}_1(t)}, \qquad (4.2)$$

$$\frac{1}{\bar{T}_1(t)}=t\left(\frac{1}{y}-\frac{1}{y+1}\right). \qquad (4.3)$$

Fig. 5 shows $\bar{T}_1$ against t for $y_1=3$. It is interesting to point out that the temperature dependence of $1/T_1$ is of opposite sense for $y_0>0$ and $y_0<0$. For $y_0<0$, we have a magnetic order in the ground state and the behavior of $1/T_1$ in Fig. 5 is similar to the one calculated for a 2D Heisenberg antiferromagnet.[15]

In actual analyses of experimental results for high T_c oxides[16] we need to consider the contribution of the other mechanisms, too. Equation (3.2-3) gives the contribution of spin fluctuation components around the antiferromagnetic wave vector Q only. Although this mechanism is expected to be predominant for the nuclei of magnetic atoms, we have additional mechanisms arising from the spin fluctuation components around $q=0$ and from the orbital currents. We may thus write

$$1/T_1=(1/T_1)_s+(1/T_1)_u+(1/T_1)_{orb}. \qquad (4.4)$$

Since the uniform susceptibilities of high T_c oxides are almost temperature independent, $(1/T_1)_u$, the contribution of spin fluctuation components around $q=0$, is expected to be almost proportional to T. $(1/T_1)_{orb}$, the orbital

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contribution is usually proportional to T. In high T_c oxides, where the relevant d bands are expected to be mainly of t_{2g} character, the orbital contribution to the relaxation may be neglected. It has been pointed out that the contribution of antiferromagnetic spin fluctuations [eq. (4.2-3)] hardly contribute to the relaxation of ^{17}O and ^{89}Y, etc., because of the symmetry of their hyperfine couplings, i.e., A_Q vanishes for these nuclei in these high T_c oxides.[17,18] Thus the relaxation rates for these nuclear spins are mostly attributed to $(1/T_1)_u$. This means that one may estimate $(1/T_1)_u$ for ^{63}Cu by using the experimental results for $^{17}(1/T_1)$ and the ratio of the hyperfine coupling constants for ^{63}Cu and ^{17}O. The experimental results of ^{63}Cu relaxation rates in various high T_c oxides have recently been analyzed in this way.[19] The deduced temperature dependence of observed $(1/T_1)_s$ is quite similar to the theoretical results in Fig. 4 with $y_0>0$. This indicates that we are in a *nearly* antiferromagnetic regime and not in a 2D-antiferromagnetic regime.

Although we do not have enough information for detailed quantitative comparison at present, we will now discuss the orders of magnitude of the ralxation rates. We estimate $\hbar\gamma_N^2 A_{hf}^2/2\pi k_B \simeq 3\times 10^6$ from $A_{hf}\simeq 220\,\mathrm{kG}/\mu_B$. The saturation value of $1/\bar{T}_1$ is ~ 2.5 for $y_0=0.01$, $y_1=3$. This value depends weakly on y_1.

The observed saturation value of $(1/T_1)_s$ is $2\times 10^3\,\mathrm{sec}^{-1}$ and $1\times 10^3\,\mathrm{sec}^{-1}$, for $La_{1.85}Sr_{0.15}CuO_4$ and $YBa_2Cu_3O_{6.91}$, respectively. From these values we evaluate $T_A\simeq 3500\sim 7500$ K, a reasonable order of magnitude as will be discussed below. The value of $T_0\sim 2000$ K as estimated in the preceding section also seems to be reasonable when we compare Fig. 4 with experiment.

Now let us make a crude estimation for the order of magnitude of the values for T_A. From the measured value of the uniform magnetic susceptibility for $La_{1.85}Sr_{0.15}CuO_4$ ($\sim 2\times 10^{-7}$ emu/g), we estimate $\chi/2\mu_B^2$ per Cu atom to be $1.2\,\mathrm{eV}^{-1}$. Assuming the band width to be $1\sim 2$ eV, or $[2\chi_0(0)\simeq 1\sim 0.5\,\mathrm{eV}^{-1}]$, we estimate the exchange enhancement factor for the uniform susceptibility $(1-\alpha_u)^{-1}$ as $1.2\sim 2.5$. Thus we have $\alpha_s/\alpha_u=\chi_0(Q)/\chi_0(0)=6\sim 1.7$, and therefore $T_A=Aq_B^2/2\chi_0(Q)\sim[1-\chi_0(0)/\chi_0(Q)]/\chi_0(Q)$ is estimated roughly to be $3000\sim 10000$ K.

Finally we emphasize again that the theory for the nearly antiferromagnetic regime seems to explain the observed nuclear spin relaxation rates. We note that an RPA calculation of $1/T_1$ based on the 2D Hubbard model with a specific electron occupation within the nearly antiferromagnetic regime was reported to give a fair explanation of the observed results.[18]

§5. Specific Heat Due to Spin Fluctuations

Specific heat due to spin fluctuations is given within the present model by[20]

$$C_m/T=-\partial^2 F_{sf}/\partial T^2,$$

$$F_{sf}=\sum_q\int_0^{\omega_c} d\omega\left\{\frac{\omega}{2}+T\log(1-e^{-\omega/T})\right\}\times\frac{3}{\pi}\frac{\Gamma_{Q+q}}{\Gamma_{Q+q}^2+\omega^2}. \tag{5.1}$$

After straightforward calculation, neglecting the contribution of the zero point fluctuations, we get

$$C_m=3N_A t\int_{y/t}^{(1+y)/t} du\left[-\left\{u^2-2u\frac{\partial y}{\partial t}+\left(\frac{\partial y}{\partial t}\right)^2\right\}\left\{\frac{1}{u}+\frac{1}{2u^2}-\psi'(u)\right\}-t\frac{\partial^2 y}{\partial t^2}\left\{\ln u-\frac{1}{2u}-\psi(u)\right\}\right]. \tag{5.2}$$

For nearly antiferromagnetic metals y becomes constant ($y\to y_0>0$) at low temperatures. Thus eq. (5.2) for small t becomes

$$C_m/T\longrightarrow\frac{1}{2}(N_A/T_0)\ln(1+1/y_0). \tag{5.3}$$

The enhancement of the linear specific heat or the effective mass is given by

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$$\frac{\gamma_m}{\gamma_0}=\frac{3}{4\pi^2\rho_a(0)T_0}\ln(1+1/y_0), \tag{5.4}$$

where $\rho_a(0)$ is the density of states at the Fermi level per magnetic atom. Taking $\rho_a(0)\sim 1\ \mathrm{eV}^{-1}$, $T_0\sim 2000$ K, $y_0\sim 0.01$, we estimate very roughly $\gamma_m/\gamma_0\sim 2$. The result of numerical calculation by using eq. (5.2) is shown in Fig. 6.

§6. Superconductivity Due to Spin Fluctuations

The possibility for the occurrence of d-wave superconductivity due to antiferromagnetic spin fluctuations was pointed out a few years ago.[21,22] Subsequently several authors have discussed this problem for the 2D-Hubbard model by using RPA type theories[23] Here we discuss the superconductivity due to antiferromagnetic spin fluctuations as discussed in the preceding sections by using, for brevity, a weak coupling theory (a generalized BCS theory). We consider the following effective interaction [24]

$$H_{\mathrm{int}}=-\frac{1}{N}\sum_{kk'q}J(q)(\sigma)_{\alpha\beta}\cdot(\sigma)_{\gamma\delta}a^{\dagger}_{k+q\alpha}a_{k\beta}a^{\dagger}_{k'-q\gamma}a_{k'\delta}. \tag{6.1}$$

From this form of interaction, matrix elements for even and odd parity pairings are obtained as

$$H_{\mathrm{pair}}=\frac{1}{2N}\sum_{\sigma}\sum_{kk'}\{V^{\mathrm{odd}}_{kk'}(a^{\dagger}_{k\sigma}a^{\dagger}_{-k\sigma}a_{-k'\sigma}a_{k'\sigma}+a^{\dagger}_{k\sigma}a^{\dagger}_{-k-\sigma}a_{-k'-\sigma}a_{k'\sigma})+V^{\mathrm{even}}_{kk'}a^{\dagger}_{k\sigma}a^{\dagger}_{-k-\sigma}a_{-k'-\sigma}a_{k'\sigma}\}, \tag{6.2}$$

with

$$V^{\mathrm{odd}}_{kk'}=-\frac{1}{4}[J(k-k')-J(k+k')],$$

$$V^{\mathrm{even}}_{kk'}=\frac{3}{4}[J(k-k')+J(k+k')].$$

For the effective interaction we use the particle-particle vertex part defined by

$$J(q)=\frac{I}{1-2I\bar{\chi}(q)}=I+2I^2\chi(q), \tag{6.3}$$

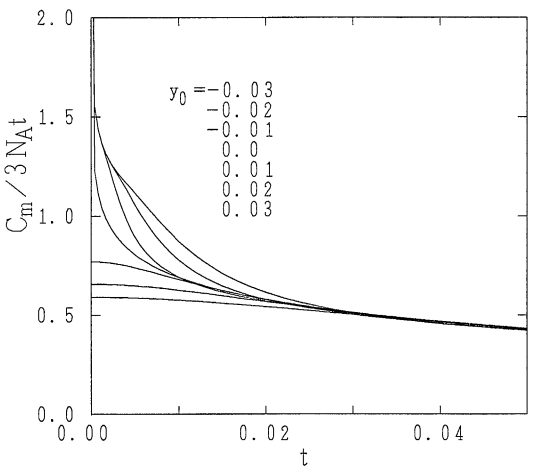


Fig. 6. Specific heat per atom due to spin fluctuations against reduced temperature for $y_1=3$.

where the intra-atomic exchange constant I and the susceptibility $\chi(q)$ have already been introduced in §2.

The transition temperature of superconductivity is expressed as

$$T_c=1.14\omega_c\exp(-1/\lambda), \tag{6.4}$$

where ω_c is the cutoff energy. The value of λ is determined as the largest eigenvalue of the linearized gap equation

$$N(0)\langle V_{kk'}\Delta(k')\rangle=-\lambda\Delta(k), \tag{6.5}$$

where $\langle\cdots\rangle$ denotes the average on the Fermi surface (line in 2D system of our concern). Here we have dropped spin indices of the gap function since we have not included spin-orbit coupling, which may be justified for the class of high T_c materials in contrast with heavy Fermion materials.

The transition temperature and the gap function just below T_c are classified according to the irreducible representations of the point group of the system. For the point group C_{4v} of the square lattice, there are five irreducible representations, A_1, A_2, B_1, B_2, and E, among which only E is two-dimensional and has odd parity (odd under the rotation of π).

A point on a Fermi line can be specified by the angle θ measured from the x-axis.

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Therefore the order parameter, $\Delta(\theta)$, is a function of θ. When the Fermi line is a circle this parametrization is nothing but the one defined by $k_x=k_F\cos\theta$ and $k_y=k_F\sin\theta$. By expanding the gap function into a Fourier series

$$\Delta(\theta)=\frac{1}{2}a_0+\sum_{n=1}^{\infty}(a_n\cos n\theta+b_n\sin n\theta), \quad (6.6)$$

it can be decomposed into the following components classified as the bases of the irreducible representations:

$$\Delta(A_1;\,\theta)=\frac{1}{2}a_0+\sum_{n=1}^{\infty}a_{4n}\cos 4n\theta,$$

$$\Delta(A_2;\,\theta)=\sum_{n=1}^{\infty}b_{4n}\sin 4n\theta,$$

$$\Delta(B_1;\,\theta)=\sum_{n=1}^{\infty}a_{4n-2}\cos(4n-2)\theta,$$

$$\Delta(B_2;\,\theta)=\sum_{n=1}^{\infty}b_{4n-2}\sin(4n-2)\theta,$$

$$\Delta(E,x;\,\theta)=\sum_{n=1}^{\infty}a_{2n-1}\cos(2n-1)\theta,$$

$$\Delta(E,y;\,\theta)=\sum_{n=1}^{\infty}b_{2n-1}\sin(2n-1)\theta. \quad (6.7)$$

Now we wish to solve the gap equation (6.5) numerically. For simplicity of calculation we use a Gaussian form around $Q=(\pi/a,\,\pi/a)$ for the susceptibility

$$\chi(q+Q)=\frac{1}{2IA}\frac{1}{\kappa_s^2}\exp\left[-\left(\frac{q}{\kappa_s}\right)^2\right], \quad (6.8)$$

which is consistent with eq. (2.3) for small q ($q<\kappa_s$). In solving the eigenvalue problem we assume a circular Fermi line for simplicity.

The eigenvalue $-\lambda$ of each irreducible representation is plotted in Fig. 7 as a function of radius of the Fermi circle k_F(k_{F0} corresponds to the half-filled case). The parameters used are typical ones: $\alpha_u=0.25$, $\kappa_s=0.354$ ($y=0.01$) and $A/a^2=0.1$. There is no attractive channel for A_1 and B_2 representations. Therefore superconductivity due to the antiferromagnetic fluctuations is not a conventional one (A_1). The largest eigenvalue occurs in the B_1 representation. It should also be noted that in a wide range of k_F the eigenvalues for B_1 and A_2 are almost degenerate. It suggests a possibility of a second phase transition in the superconducting region.

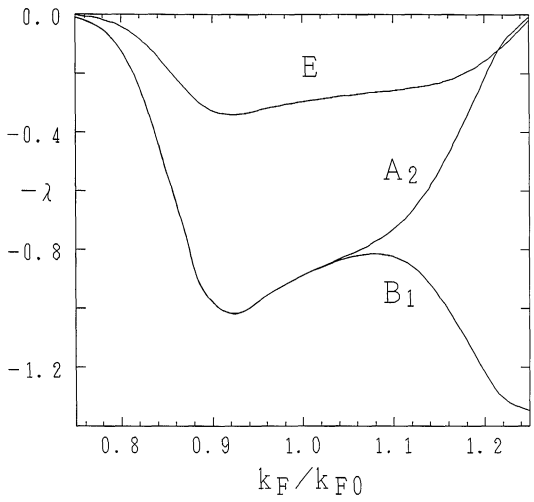
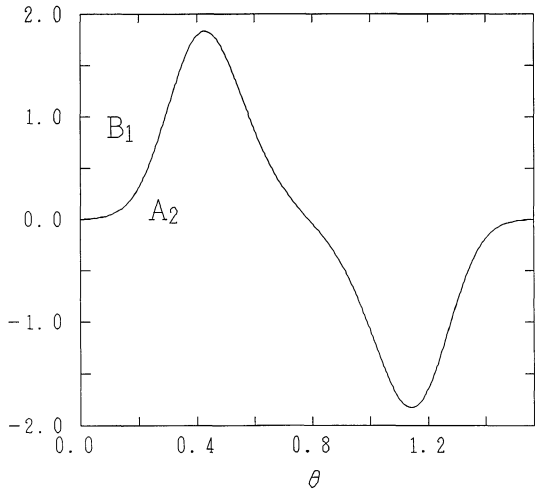


Fig. 7. Eigenvalue of each irreducible representation as a function of the radius of the Fermi circle. There is no attractive channel for A_1 and B_2 irreducible representations.

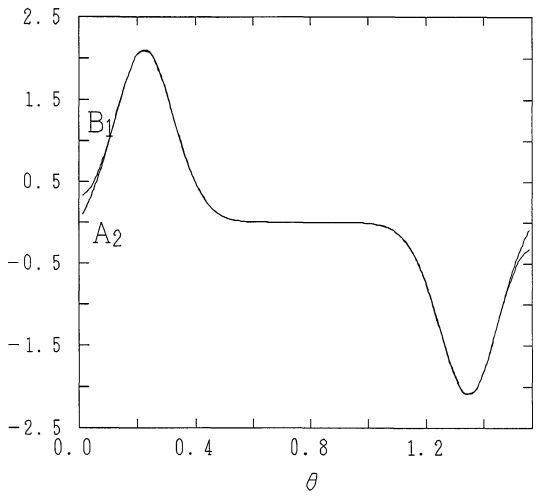
Basis functions of B_1 and A_2 irreducible representations have nodes at $\theta=\pm\pi/4$ and $\pm3\pi/4$. To use the sharp peak of $\chi(q)$ at $q=Q$ as the source of an attractive interaction, it is necessary for the gap function to have sign change at these points. Recent neutron experiments report that the antiferromagnetic fluctuations are not modified by the presence of the superconductivity.[13] This result is naturally understood if the order parameter is of a B_1 type or a mixture of B_1 and A_2. Examples of the eigenfunctions are shown in Fig. 8a ($k_F/k_{F0}=0.95$) and 8b ($k_F/k_{F0}=1.05$). Although the symmetries are different, the form of the eigenfunctions are very close when their eigenvalues are almost degenerate. Another interesting feature is that the gap function is small in a wide range of θ when k_F is large. This fact can be also traced back to the sharp peak structure of $\chi(q)$.

Finally we would like to estimate the transition temperature. For this purpose we need to take a proper value for the energy cutoff, ω_c, considering the frequency dependence of the dynamical susceptibility. Here we take $\omega_c=\Gamma_{Q+\kappa_s}=4\pi T_0 y$ as a representative value: $\omega_c=250$ K for $T_0=2000$ K and $y=0.01$. Close to the half-filling ($0.9<k_F/k_{F0}<1.1$), λ is in the range of $0.8\sim1.0$. As an estimation of T_c we get 80 K ~ 110 K. Similar value (110

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(a)



(b)

Fig. 8. Form of the order parameter for B_1 and A_2 representations ($0 \leq \theta \leq \pi/2$). (a) $k_F/k_{F0}=0.95$. For this parameter the two functions are not distinguishable. (b) $k_F/k_{F0}=1.05$.

K $\sim$ 140 K) is obtained for another choice of y($y=0.02$). These numbers are of the right order of magnitude, although we should keep in mind the crudeness of this estimation. To make a more reliable estimation we need to evaluate A or T_A more accurately possibly by neutron scattering experiments, since the eigenvalue scales with $\alpha_u a^2/A$.

§7. Conclusions and Discussions

We have seen in the preceding sections that the electrical resistivity and the nuclear spin-lattice relaxation rate in the normal state of high T_c oxides are fairly well explained as due to the spin fluctuations in nearly antiferromagnetic two-dimensional itinerant electron sytems described by the SCR theory. The same spin fluctuations are shown to give rise to the superconductivity of B_{1g} or A_{2g} symmetry below the critical temperature T_c which is estimated to be of the right order of magnitude. We emphasize that the present theory contains only a small number of parameters to specify the spin fluctuations and their values are estimated here from the experimental results in the normal state.

Although the present approach is from the side of weak correlation, our use of experimental results in evaluating the values for the parameters may effectively take into account some of the many body renormalization effects. It is highly desired that more precise observations of spin fluctuations are made and the values for the parameters are determined from neutron inelastic scattering experiments, etc. in future.

There are many more experimental results to be explained. Some of them will be explained within this model while many others may require further refinements of the model. This model seems at least to provide us with a good starting point to describe the physical properties of high T_c oxides.

Of course it is quite important and necessary to understand the physics of the continuous series of these doped oxides from the mother insulators to high T_c and normal metals beyond high T_c. The approaches from the strong correlation limit should certainly cover the range of small doping, although to be realistic additional consideration of the Anderson localization effects due to random potentials is required. The normal metal regime should be approached better from the weak correlation limit. We expect the approaches from both sides eventually come into a smooth contact. The point is which approach does contain essential concepts in a more direct way and is more convenient in each respective regime.

In this respect we should like to emphasize here that the present analysis seems to indicate

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that the spin correlations in high T_c oxides are in the nearly antiferromagnetic regime and thus are hardly described by the local moment picture or a 2D Heisenberg model with antiferromagnetic couplings. In other words the nature of spin fluctuations in the high T_c oxides seems to be significantly different from that of their mother insulators, although we still need more investigations both experimentally and theoretically before drawing definite conclusions.

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